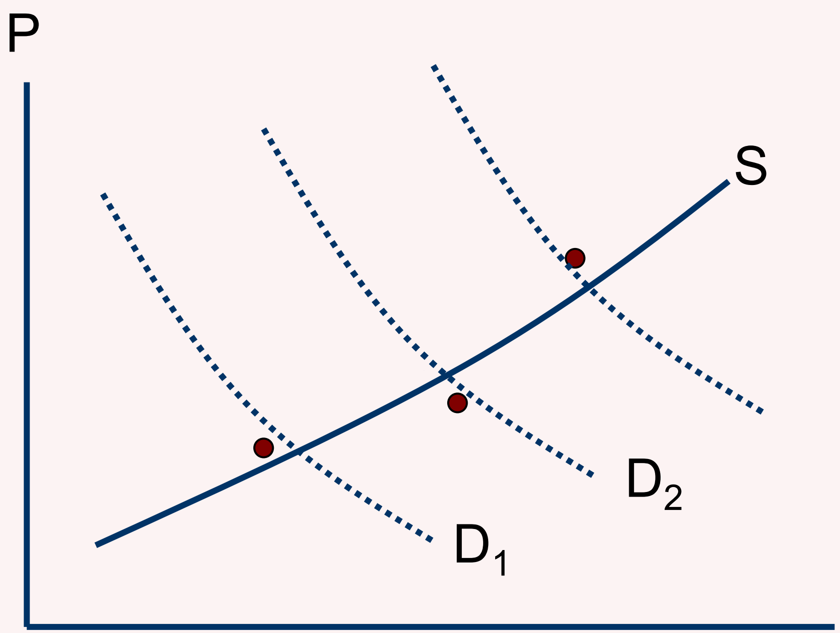
GY457
Applied Urban and Real Estate Economics

Block II—**Topic 3:**

Identifying the Housing Supply and Demand Curve

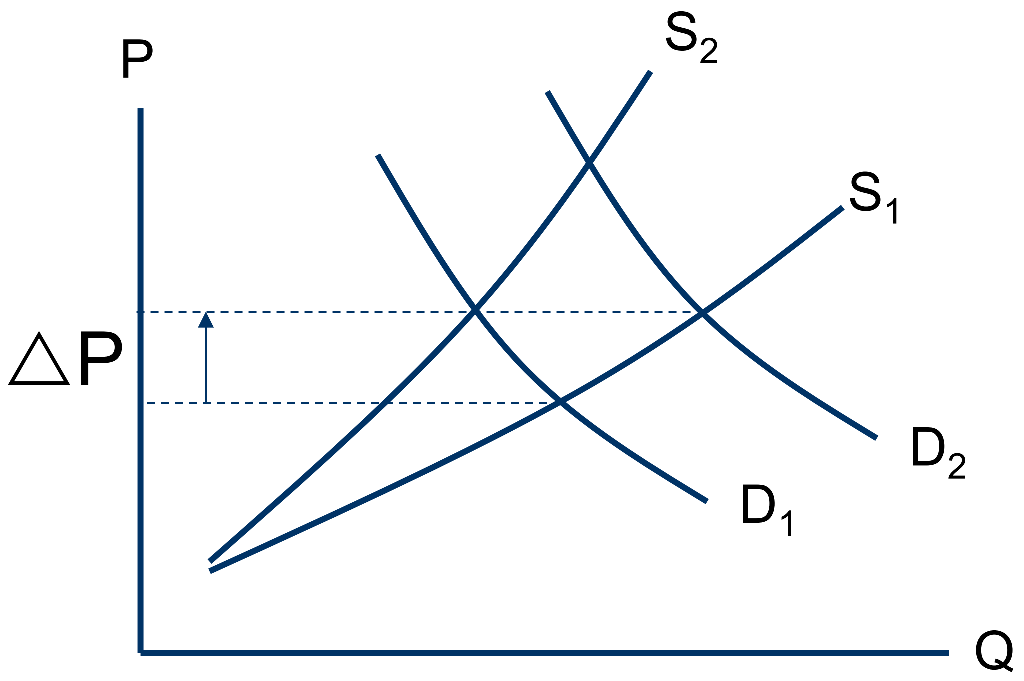


Recap of last two lectures...

- Looked at role of government
  - Local public goods: centralised vs. decentralised provision
  - Tiebout sorting & Tiebout hypothesis
  - Fiscal variables capitalized into house prices
    - Oates (1969): 2/3 capitalized
    - Later studies: partial to full capitalization
  - Land use regulation
    - Welfare economic justification: market failures
    - But also: Private and social costs & unintended consequences

Changes in house prices…

… can be result of shift of demand or supply curve …



Discuss…

- ⇨ Is house price capitalization of local services and taxes a demand or supply side effect?

- ⇨ What factors could affect extent of house price capitalization?

Task for this lecture

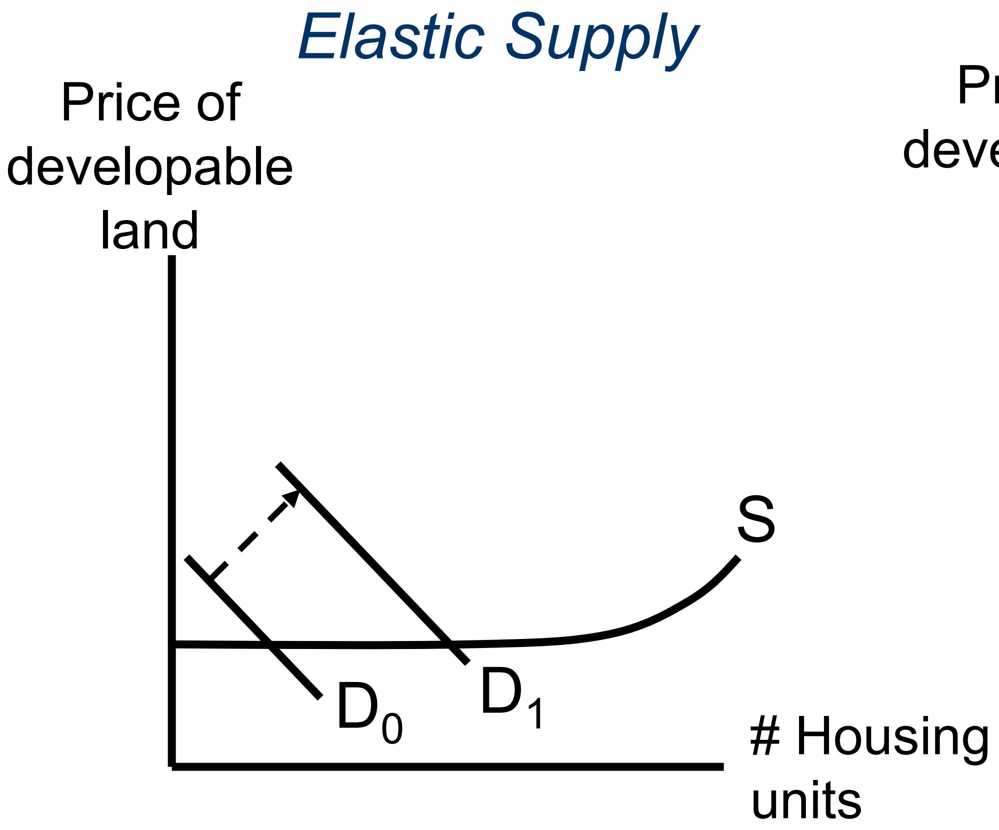
- Answer two questions
  - How sensitive does supply react to changes in house prices? *(slope of supply curve)*
  - How sensitive does demand react to changes in house prices? *(slope of demand curve)*

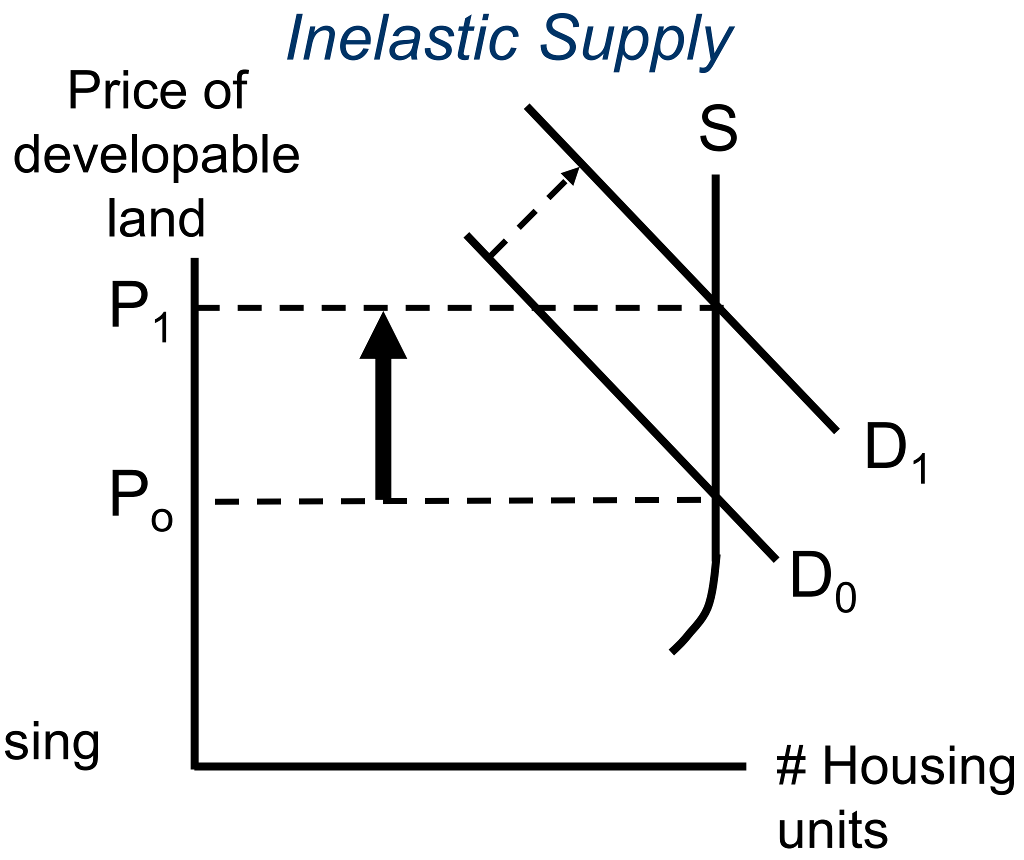
Why important?

- Real estate practitioners and investors interested in questions such as
  - What is the effect of a new railway station on house prices in the surrounding area?
- Policy makers interested in questions such as
  - What is the effect of financial aid for local beneficiaries?

Why important?

- Consider shift of demand curve…





Contents—Lecture 3

1. The Identification Problem
2. Estimating the Supply Price Elasticity
3. Estimating the Demand Price Elasticity
4. Summary of Findings and Implications

Contents—Lecture 3

Starting point

- Goal
  - Estimate effect of change in house price on housing supply and demand
- Initially available data
  - House price index (time-series)
  - New housing permits in each period

Consider following setting…

1. One local housing market (abstract from interdependencies across markets)

2. Abstract from heterogeneity in housing stock (housing units are comparable)

3. Market is in equilibrium in each time period (demand = supply)

How can we estimate supply and demand curve?

- Idea: Use <u>regression technique</u>
- A quick "Econometrics 101" refresher…

Regression technique & Ordinary Least Squares (OLS)

- Regression technique allows us to assess causal link between variable *X* and outcome *Y*
  - How much does change in *X* affect *Y*?

- Linear regression analysis tries to answer question by linear specification of relation between *X* and *Y*

$$y = \alpha + \beta x + \gamma c + \varepsilon$$

y…dependent (outcome) variable (individual observation)
x…explanatory variable of interest
c…vector of control variables
ε…error term

α…constant
β…parameter we want to estimate
γ…parameters of control variables

What exactly is OLS?

- Given Y and X, find values of β and α that best 'fit' data
  - Want to draw linear approximation (constant + slope)
  - How choose line? Minimize distance between fitted and actual Y… This is OLS!
- What do we find? Coefficient β expresses change in Y as result of change in X
  - More precisely covariance of (X,Y) over variance of X
  - Measures strength of relation between X and Y, deflated by variation in X

How can we estimate supply and demand curve?

- Idea: Use OLS
  - Supply equation:

$$Q_t^S = \alpha_0 + \alpha_1 P_t + \varepsilon_t \qquad (1.1)$$

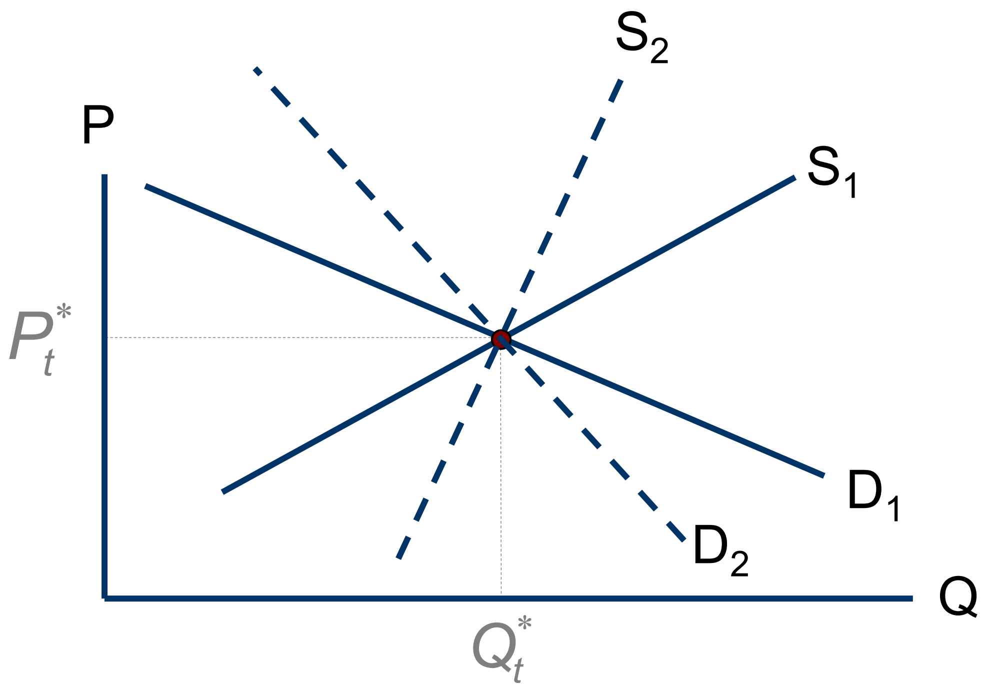
  - Demand equation:

$$Q_t^D = \beta_0 + \beta_1 P_t + \mu_t \qquad (1.2)$$

⇨ Can we estimate these equations?

Identification problem

- At each time period we only observe equilibrium
  - Market price
  - Quantity of housing units sold



Why unidentified?

- Rearrange equations (1.1) and (1.2) assuming $Q_t^D = Q_t^S = Q_t$ and solve for P_t and Q_t to get "reduced form" equations:

$$P_t = \frac{(\mu_t - \varepsilon_t) + (\beta_0 - \alpha_0)}{(\alpha_1 - \beta_1)} \qquad (2.1)$$

$$Q_t = \frac{\alpha_1\beta_0 + \alpha_1\mu_t - \alpha_0\beta_1 - \beta_1\varepsilon_t}{(\alpha_1 - \beta_1)} \qquad (2.2)$$

How can we identify the slopes of the supply and demand curves?

- Need further information…
- Assume we also know one (exogenous) **“demand shifter”** – Z_t
  - Variable Z_t only shifts demand but not supply curve
- Possible demand shifter? *(Discuss)*
  - Average HH income in local housing market
    - But could be argued to also affect supply side via construction sector wages &
    - HHs sort across space – local income could be correlated with variables that shift supply curve
  - Better alternative? ‘Labor demand shock’-measure (shift-share approach; ‘Bartik-instrument’)

New system of equations

- Supply equation:

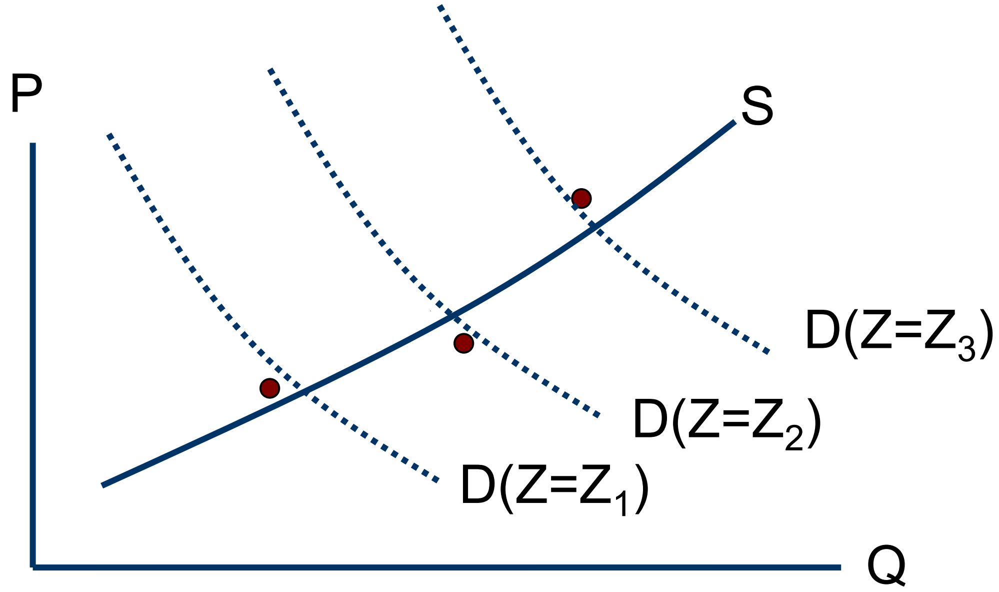
$$Q_t^S = \alpha_0 + \alpha_1 P_t + \varepsilon_t \qquad (3.1)$$

- Demand equation:

$$Q_t^D = \beta_0 + \beta_1 P_t + \beta_2 Z_t +_t \mu_t \qquad (3.2)$$

⇨Which curve is now identified? *(Discuss)*

What is effect of increase in Z_t over time?



⇨ *Supply equation is identified!*

Consistent estimation of supply parameters

- Variable Z_t only affects demand but not supply
  - Known as "identifying assumption"
- Variable Z_t is so called "instrumental variable"
  - Z_t affects P_t but is uncorrelated with error term ε_t (no independent effect on supply of housing)

⇨ *How to estimate in practical terms?*

Can't use **OLS**…why exactly? *(Discuss)*

- Crucial assumption of OLS: RHVs are all exogenous
- Here…
  - Simultaneous equation
  - Problem of reverse causation and endogeneity

⇨ *Estimates of α_1 and β_1 are biased (wrong)*

⇨ *How can we fix this problem?*

Use Two-Stage-Least-Squares (TSLS or 2SLS) estimator

- First stage
  - Estimate P_t as function of exogenous variables (here Z_t)

$$P_t = \gamma_0 + \gamma_1 Z_t + v_t \longrightarrow \hat{P}_t$$

- Second stage
  - Estimate Q_t using predicted value of endogenous variable from first stage

$$Q_t = \delta_0 + \delta_1 \hat{P}_t + \varpi_t$$

unbiased

(Note: δ_1 is inverse of slope of supply curve)

▷ In Stata

Intuition behind **TSLS**-approach

- First, predict variation in P using variation in Z
- Relate variation in P that is ‘explained’ by variation in Z to variation in Q
  - Why does this help? Crucial assumption is: Z only affects demand (not supply)
  - So, prediction of P using Z does not contain movements in supply that contaminates estimates
  - Idea: Keep S fixed, move D, get slope of S (linear approximation)

Rules of thumb for identification?

- Order condition
  - Number of predetermined/exogenous variables [Z_t] excluded from equation must be greater than/equal to number of included endogenous variables [P_t or Q_t] minus 1
- Power of instrument
  - Predetermined/exogenous variables [Z_t] must be good predictors of endogenous variables (explain a large fraction of variance in P_t)

Rules of thumb for identification? (Cont.)

- Exogeneity
  - Variable used as instrument [Z_t] is not related to error term in system of equations (can be treated as a shifter; not a result of the system or 'proxy' for something unobserved)
- Exclusion restriction
  - Exogenous variable should not directly affect outcome of interest (only affects it via changes in the endogenous/instrumented variable)

Another issue:
Levels or changes?

- *Quantity* is usually measured as number of newly built housing units or number of permits or starts
  - But really ΔQ rather than Q
  - Flow-measure not stock-measure
- ΔQ is then usually regressed on prices in levels

⇨ Problem: While starts are stationary*, prices are not!

* A stationary time series is one whose statistical properties such as mean, variance, autocorrelation, etc. are all constant over time.

Another issue: Levels or changes? (Cont.)

- However, price *changes* are stationary!

⇨ Estimate *change* in quantity (starts) as a function of *changes* in other measures (or percentage change in stock as function of percentage change in prices & other covariates)

▷ Why not levels or log of levels?

"Correct model"

Supply Equation:

$$\Delta Q_t^S = \alpha_0 + \alpha_1 \Delta \hat{P}_t + \alpha_2 \left(\Delta Supply\ Shifters_t \right) + \varepsilon_t \qquad (4.1)$$

Predicted value from 1st stage: Estimated using instrumental variables (=predetermined supply and demand shifters)

Demand Equation:

$$\Delta Q_t^D = \beta_0 + \beta_1 \Delta \hat{P}_t + \beta_2 \left(\Delta Demand\ Shifters_t \right) + \mu_t \qquad (4.2)$$

▷ Some test-statistics

Empirical evidence?

- Various studies attempted to estimate structural models with supply and demand equations
- Attempt to obtain sensible and unbiased estimates of
  - Supply price elasticity
  - Demand price elasticity

⇨ Consider estimates of (structural) supply equations first…

Contents—Lecture 3

Starting point: What exactly is a price elasticity?

- A measure of the sensitivity of supply and demand to changes in price
- The price elasticity can be expressed as:

$$\mathrm{E}^{S} = \frac{\Delta Q^{S} / Q^{S}}{\Delta P / P} \qquad \mathrm{E}^{D} = \frac{\Delta Q^{D} / Q^{D}}{\Delta P / P}$$

⇨ If house prices increase by 1 percent, then housing supply (demand) increases (decreases) by X percent

Estimates of supply price elasticities

- Poterba (1984)
  - Look at single-family housing in US
  - Quarterly data from 1974-1982
  - Q_t: Net investment in SF-structures *(flow)*
  - P_t: Real house price *(level)*
  - Using lagged variables as instruments
- Estimated supply elasticity [of construction]: **0.5 – 2.3** (depending on specification)
  - 10% increase in real price increases net investment by 5-23%

Estimates of supply price elasticities (Cont.)

- Topel and Rosen (1988)
  - Look at single-family housing in US (1963-1983)
    - Q_t: Single family housing *starts* per quarter *(flow)*
    - P_t: Real hedonic price index
    - Additional supply and demand shifters
  - Estimates
    - Short-run supply elasticity [of constr.] (1 quarter): **1.0**
    - Long-run supply elasticity [of constr.]: **3.0** (most adj. completed within one year)

⇨ Housing supply is much more elastic in long-run than in short-run

Critique?

- Earlier studies estimate new housing supply (*change* in housing stock) as function of housing prices (*level* measure)

⇨ Mayer and Somerville (2000)

  - ***Change*** **in housing stock should be explained by** ***change*** **in prices!** (or percentage change in stock by percentage change in prices)

  - Why not levels or log of levels?

Using first differences

- Mayer and Somerville (2000)
  - Show starts and price *changes* are stationary
  - Use first difference (△) specification
    - Starts interpreted as △Q
    - Change in price is △P
  - *Also* use changes in cost variables
  - Elasticity of supply [of stock] is 0.08
    - 10% increase in price increases housing *stock* by 0.8% *spread over 4 quarters*
  - Short-term elasticity of *starts* (one q) is 6.3
    - 10% increase in price increases quarterly starts by 63% immediately

Summary: Comparison of 'macro' elasticities

	LR elasticity of supply (*stock*) (dQ/Q)/(dP/P)	**SR elasticity of starts (1quarter)** (d△Q/△Q)/(dP/P)	**LR elasticity of starts** (d△Q/△Q)/(dP/P)
Poterba		0.5-2.3*	
Topel & Rosen		1.0*	3.0* (most of adj. within 1 yr)
Mayer & Som.	0.08† (assume temporary increase in starts, 4q)	6.3† (quarterly starts, contemp.) t-1: 6.3, t-2: 4.3, t-3: 1.5	3.7† (annual starts; over 4 q) (effect dissipates, but aggregated effect on supply larger)
Notes: *… traditional approach (regress starts on prices), †…first differences (i.e., regress starts on △P); Q…stock, △Q…starts, P…price			

▷ How does 0.08 'map' into 3.7 and 6.3?

Critique?

- All these elasticity estimates are at country level (US)
- BUT: Supply price elasticity may vary across space
  - Rural locations with plenty of open land likely have elastic supply
  - Urban locations with little open land likely have inelastic supply

⇨ Why exactly? *(Discuss)*

Theoretical justifications for proposition? *(Discuss)*

1. Physical constraints (no land to build, costly to build, land as exhaustible resource)

2. In urban locations owners of developed land are politically more powerful ⇨ limit future construction via land use restrictions/zoning *(Hilber and Robert-Nicoud 2013)*

3. Real options argument (last parcel(s) of open land extremely valuable)

Derived testable prediction

⇨ Locations with less undeveloped land have more inelastic housing supply

Empirical evidence?

- Hilber and Mayer (2009)
  - Look at 208 communities in Commonwealth of Massachusetts (US)
  - Split into two equal sized groups based on percentage of land that is undeveloped in 1984
  - Estimate structural model with first difference specification
  - Expect more inelastic supply in places with less undeveloped land

Land supply elasticity for more and less developed municipalities

Dependent variable: SF permits (1990-94) per 1990 housing units

$$Equation: \frac{\Delta Q}{Q} = \beta_1 + \beta_2 \frac{\Delta P}{P} + \beta_3(Controls) + \varepsilon$$

Specification	**(1)**	
Explanatory Variable	Less Land	More Land
Percent change in house prices, 1990-1994 (instrumented)	**.014** **(.055)**	**.16 **** **(.079)**
Constant	**.043**** **(.0056)**	**.064**** **(.0086)**

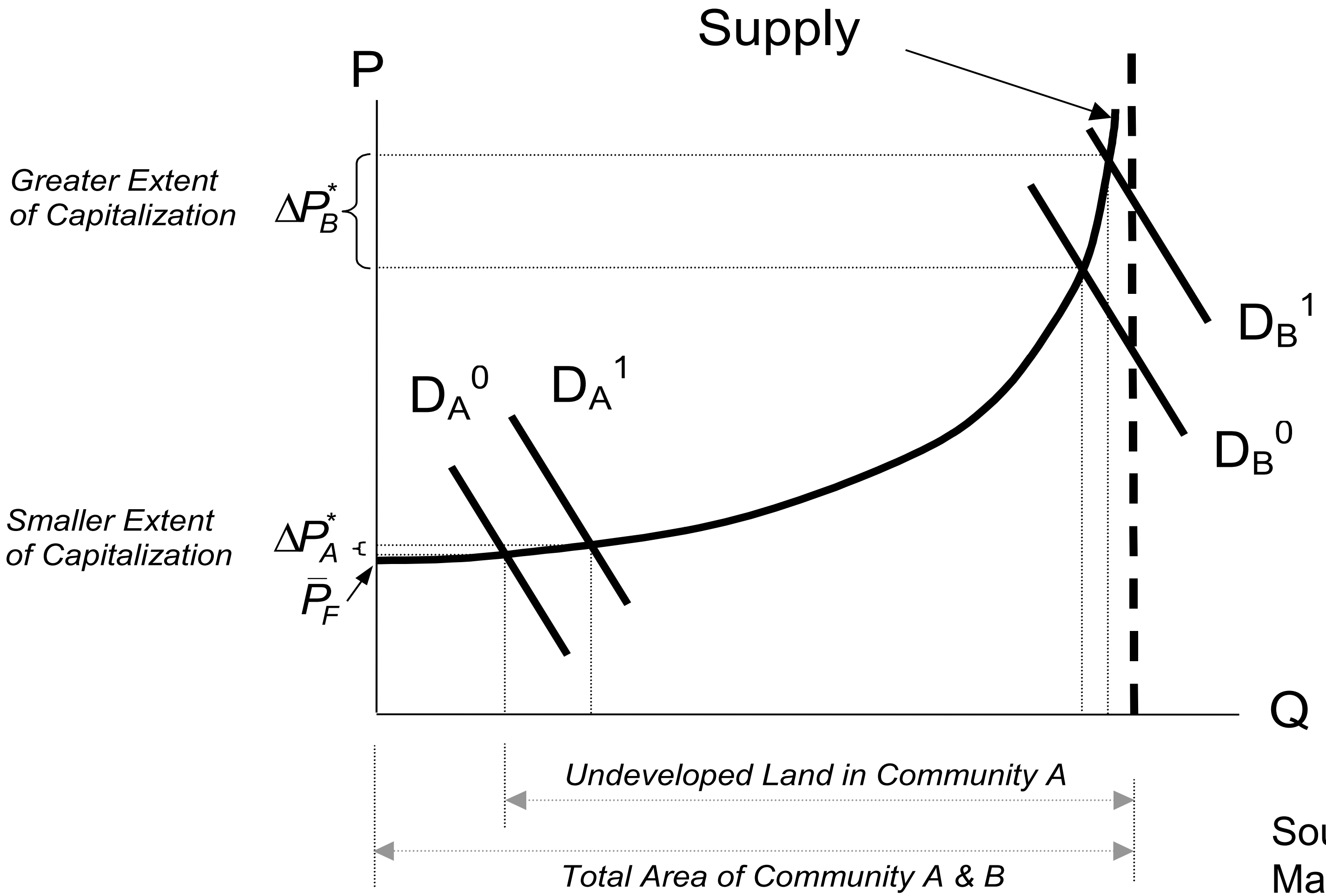
Preliminary conclusion

- Locations with less undeveloped land have more inelastic supply
  - Supply price elasticity in urban locations: 0.014
  - Supply price elasticity in rural locations: 0.16

Implications

- If supply is more inelastic in more developed locations…
  - House price capitalization of fiscal variables and local amenities should be greater in places with less undeveloped land

Land availability and house price capitalization



Source: Hilber & Mayer (2008)

Hilber & Mayer (2009) —Approach

- Estimate the effect of school spending and other local amenities on house prices for both groups of communities
- Look at changes to control for omitted variables
- Use instruments to identify Δ Spending and Δ Housing Stock in order to control for the endogeneity problem

Does the land supply elasticity affect the extent of house price capitalization?

Dependent variable: % Change in house prices 1990-1994

Specification	Split on Land Availability	
Explanatory Variable	Less Land	More Land
Percent change in school spending, 90-94	**.33 ** (.12)**	**.099 (.11)**
Percent change in non-school spending, 90-94	**.075 (.086)**	**.017 (.061)**
Math and reading MEAP test score, 90 (x 10^3)	.14 ** (.029)	.11 ** (.031)
Dummy variable, in Boston metro area	.097 ** (.012)	.074 ** (.011)
Dummy variable, in Boston suburban ring	.11 ** (.022)	.036 ** (.0091)
Single family permits, 90-94, per 90 housing units	-.74 ** (.22)	-.11 (.17)

Other more recent 'local' supply price elasticities: Saiz (2010)

- Saiz (2010) estimates (inverse) supply price elasticity for large sample of metro areas (MSAs) in US
  - Inverse supply elasticity β_j as $\partial lnP_j / \partial lnH_j$
    - H_j refers to # of HHs in MSA *j* rather than housing stock (Q_j)
    - Recall: $\partial lnP_j \approx \Delta P/P$ & $\partial lnH_j \approx \Delta H/H$
    - Estimate of $\varepsilon_j = 1/\ \beta_j$ is <u>very</u> long-run: 1970-2000
    - Average elasticity is around 1.7 (varies from 0.6 to 5.5)
    - Elasticity measured for entire MSAs so including edge of cities, where housing can typically be easily added in US

Local supply elasticities: Baum-Snow & Han (2024)

- Estimate housing supply elasticities for US at very disaggregated level
  - Also long-run: 2000-2010
  - Estimate of average floorspace elasticity across all urban neighborhoods in US: 0.5
  - Estimate of average unit elasticity: 0.3
  - In inner city (built-up) neighborhoods much lower (closer to zero)
  - New construction explains about 50% of unit supply response – rest to be explained by teardowns & renovations

And in the UK?
Hilber and Mense (2025)

- Follow Saiz (2010) and estimate very long-run supply price elasticity for LAs (& regions) in England
  - Inverse supply elasticity β_j as $\frac{\partial lnP_j}{\partial lnH_j}$
  - Estimate of $\varepsilon_j = 1/\ \beta_j$ is very long-run: 1981-2011
  - Average elasticity is around 1.16 [↔ 1.7 for US]
  - Varies from 0.15 (City of London) to 3.6 (St. Edmundsbury, East of England) [↔ 0.6 – 5.5 for US]
  - Considerably less elastic than estimates by Saiz for US

Summary: Comparison of 'local' elasticities

	LR (unit) elasticity of supply (*stock*) (dQ/Q)/(dP/P)		
	Time horizon	**Average / Range**	**Geography**
Hilber & Mayer (2009)	4 years (1990-1994)	Urban: 0.014-0.13† Suburban: 0.16-0.18†	Municipalities in Massachusetts (urban: near Boston)
Saiz (2010)	30 years (1970-2000)	Average: 1.7 (0.6-5.5)	US metro areas
Baum-Snow & Han (2024)	10 years (2000-2010)	Average: 0.29 Greater variation within than between MSAs	Urban neighborhoods (Census tracts)
Hilber & Mense (2025)	30 years (1981-2011)	Average: 1.16 (0.15-3.6)	Local Authorities in England

Notes: † This compares to 0.08 (over 1 year) in Mayer and Somerville (2000). Mayer and Somerville (2000) look at effect of one-time price shock on subsequent construction. Other studies look at effect of price change over certain time period on change in stock during same time period.

Conclusions

- Extent of house price capitalization is much greater in places with little undeveloped land (inelastic supply)

⇨ More incentives for homeowners to invest in local infrastructure or good schools in more developed places

⇨ What about the demand price elasticity?

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Estimating the demand price elasticity

- Similar approach
  - Structural model and use of supply shifters to identify price variable
- Most studies estimate direct price elasticity of **individual** demand for housing
- Findings of various studies are more consistent than supply-side studies
  - Estimated demand price elasticity typically lies between -0.1 and -1.0 (in UK around -0.4, see Ermisch *et al.* 1996)
  - Credit-constrained HHs have more inelastic reaction to changes in prices (Rosenthal *et al.* 1991)

▷ Details

Contents—Lecture 3

Summary of main findings

1. Housing **supply** is much **more price elastic in long-run** than in short-run *(part of supply response is lagged)*

2. Housing **supply** is **more price elastic in places with plenty of open land** (and laxer land use regulation)

3. The **demand price elasticity** is **similar across locations** but is **lower for credit-constrained households**

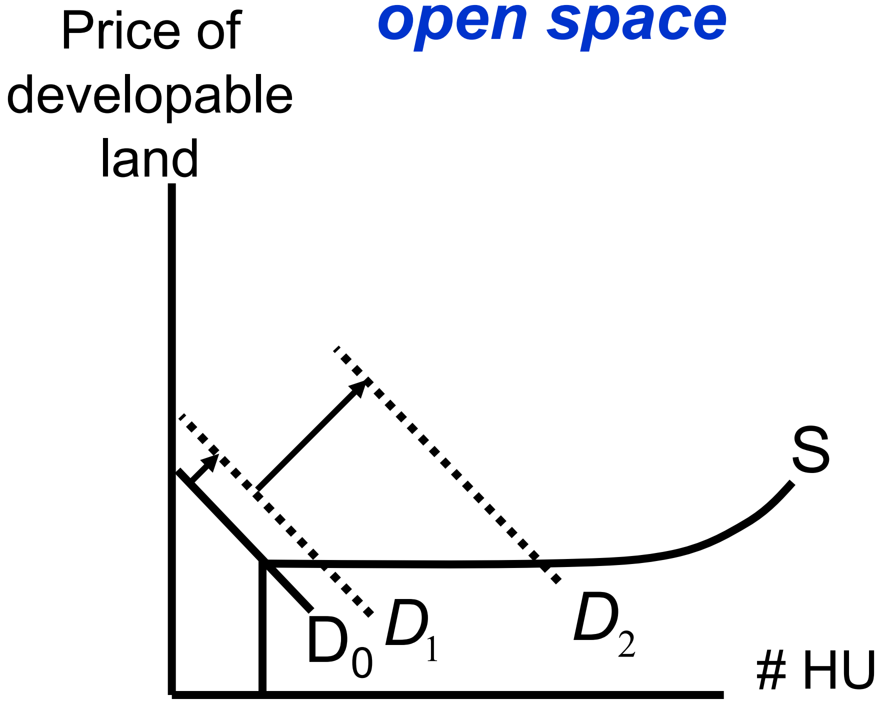
Implications for real estate markets

- Demand volatility (cyclicality) should lead to more pronounced house price volatility (cyclicality) in locations with inelastic supply of new housing
- Why?

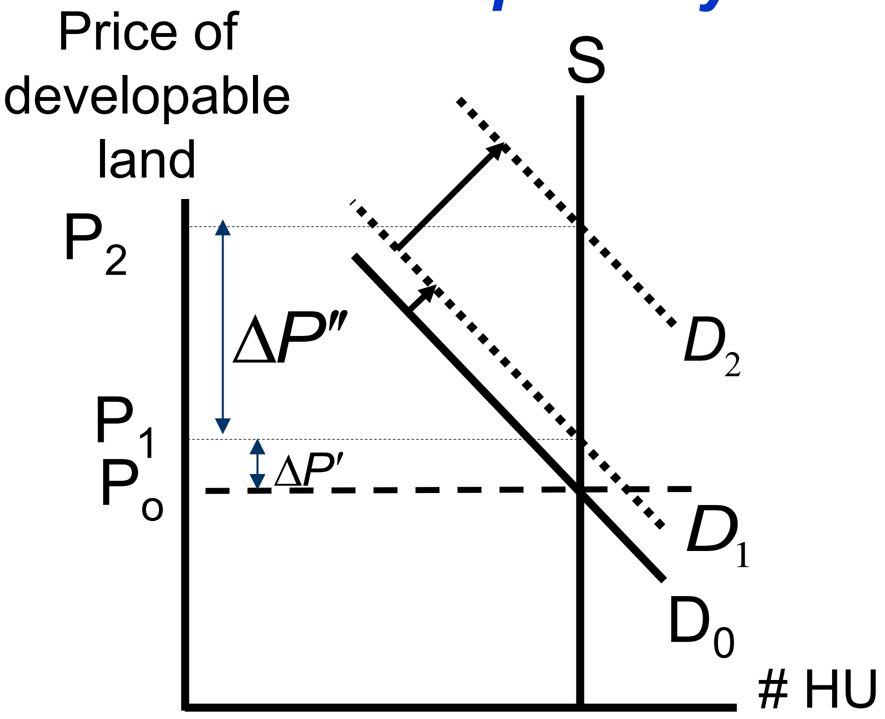
Why?

- Effect of demand volatility (cyclicality) on land and house prices…

City with plenty of open space



Completely developed city



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Implications for real estate practitioners and investors

- Return on real estate project investment depends (among other things) on **long-run supply price elasticity**
  - How easily can project be replicated?

- Other relevant question to ask: Are expected changes in demand fundamentals **already anticipated** and priced into value of land?

Policy implications?
(Discuss)

- If supply is price inelastic: Do not help places, help people
  - But also depends on HH mobility
  - In extreme: In world with income segregation & perfectly immobile HHs: no difference
- Examples of likely ineffective policies
  - Central government/federal grants to poorer regions (Hilber *et al.* 2011)
  - Inter-jurisdictional financial equalization
  - Housing aid for centre city dwellers
  - Housing subsidies (Hilber & Turner 2014, Carozzi *et al.* 2024)

Preparations for *Topic 3* – Seminar 3 (week 5)

- **Problem set (case study) with sub-group presentations and discussion** (building on Lecture 3)
  - Topic: Estimating the housing supply price elasticity for UK regions
  - Details: See special handout
  - Relevant key readings: See handout

Preparations for *Topic 4*—Seminar 4 (week 8)

- Small group presentations with discussion
  - Topic: Housing market dynamics around the globe – facts and determinants
  - Details: See special handout (on Moodle)
  - Aim: Illustrate housing market dynamics in different markets and try to explain…can we make sense of empirical facts?

Addons

In Stata...

ivregress estimator depvar [varlist1] (varlist2 = varlist_iv) [if] [in] [weight] [, options]

Example:

Estimator

Dependent variable

Exogenous explanatory variables

Endogenous explanatory variable

ivregress 2sls dquantity dcontrols (dprice = dinstrument), first

Tells STATA to report First-Stage results including test-statistics

[Excluded] **Instrument** (e.g., Bartik-instrument)

Addon: Why not just using logs?

- A convenient way of identifying price elasticities would be to measure Q and P in natural logs
  - Then the estimated coefficient could be directly interpreted as elasticity
- Problem: Housing stock and real price of housing [or natural logs] are neither stationary nor cointegrated* (Mayer & Somerville, 2000, JUE, p. 98, FN16)

*Cointegration is a statistical method used to test the correlation between two or more non-stationary time series in the long run. Two sets of variables are cointegrated if a linear combination of those variables has a lower order of integration. [They have a common trend that combines them in the LR.]

Addon: Why not just using logs?—*cont.*

- However, first differencing of
$$lnQ = \alpha + \beta lnP$$
yields
$$dlnQ = \beta dlnP$$
- While P and Q are not cointegrated and lnP and lnQ are likely also not cointegrated, dlnQ and dlnP are likely cointegrated. Why? Recall: $\partial lnP \approx dP/P$ & $\partial lnQ \approx dQ/Q$ and dP/P and dQ/Q are both stationary and cointegrated

Tests for validity of instruments?

- Overidentification tests *(Hansen-Sargan tests)*
  - H_0: Instruments are valid instruments
  - Implemented in standard econometrics software
- Tests for endogeneity of instruments *(Durbin-Wu-Hausman tests; Wu-Hausman tests; C-statistics or diff-in-Sargan-statistics)*
  - H_0: OLS would yield consistent estimates
  - Tests for all or subset of instruments
  - Also implemented in standard software

More on testing for validity of instruments

- Some economists not ‘enthusiastic’ about these tests due to (implicit) assumptions
  - Take over-id test: Assumes that at least sub-set of ‘correct’ instruments exist among those used
  - Failure to reject Null does not mean *all* instruments are valid
- Other important feature of instruments is that they must be good predictors of endogenous variable
  - Strength of relation can be gauged by looking at 1st-stage F-test (better K-P F-stat or A-P F-stat)
  - Displayed in most statistics-software (e.g., STATA)

Addon: How to interpret different supply price elasticity estimates?

- Using data from Mayer & Somerville (2000) [M&S]:
  - Q=59.9m owner-occupied housing units (stock)
  - M&S **estimate for elasticity of stock***:* 0.08
  - 10% increase in P incr. stock by 0.8% (=+0.48m units)
- Mayer and Somerville estimate of **elasticity of construction** over 4q is 3.7
  - Starts p.a. in US is 2.2% of stock (p. 89 M&S) = 1.3m units
  - 10% increase in P increases construction over 1 year by 37%
  - 37% of 1.3m = +0.48m units
  - Of those 0.48m units, majority are produced within first 2q after shock according to M&S
  - But constr. (in absolute terms) increases more over whole year than over just 1 quarter

Addon: Interpretation of Mayer and Somerville (2000) (Cont.)

- "A 10% rise in real HPs leads to an 0.8% incr. in the housing stock; this is accomplished by an immediate 63% incr. in quarterly starts. Over a year, annual starts increase by a total of 37%. This analysis highlights **the difference between a housing supply elasticity and the starts elasticity found in the existing literature**. In our modeling framework, a one-time increase in housing prices **leads to a temporary rather than permanent increase in new construction, yielding a finite increase in the stock of housing**." (p.105)
- "A 1% increase in prices in a given quarter causes a 6.3% increase in starts in the same quarter. This is approximately six times the estimate from […] Topel and Rosen's short-run elasticity. Over a year, **a one-time 1% increase in prices causes** annual starts to rise 3.7%. This is only slightly higher than Topel and Rosen's long-run estimate." (p. 102)

Addon: How to get to the 0.8% and the 63%.

- Stock in 1,000 is 59851
- Starts per quarter are 259.9 (seasonally adj.)
- Price is $84,154
- Coefficients in Table 3, Column (1) [page 100] on $\triangle P_t$, $\triangle P_{t-1}$, $\triangle P_{t-2}$, and $\triangle P_{t-3}$ are 0.0194, 0.0196, 0.0134, and 0.0047
- 1% increase in Price is $841.54
- This leads to increase in stock in t of 0.0194*841.54=16.33 in t, 0.0196*841.54=16.49 in t+1, 0.0134*841.54=11.28 in t+2 and 0.0047*841.54=3.96 in t+3 (a total of 48.05)
- 48.05 / 59851 = 0.080%
- 16.49 / 259.9 =6.3%

Demand side studies

- Europe: Ermisch *et al.* (1996)
  - Estimate demand price elasticity at around -0.4 using data for six UK regions
- United States: Various studies
  - Rosenthal *et al.* (1991)
  - Rapaport (1997)
  - Ioannides and Zabel (2002)

Demand side studies (Cont.)

- Rosenthal *et al.* (1991)
  - Look at 1981 American Housing Survey (75,000 households)
  - Estimate demand price elasticity based on *all* households: -0.92
  - Based on *credit constrained* HH only: between -0.38 and -0.54

⇨ Credit constrained households have more inelastic reaction to changes in prices

Demand side studies (Cont.)

- Rapaport (1997)
  - HH simultaneously choose (1) location, (2) housing tenure and (3) quantity of housing
  - Estimate simultaneous equation model for Florida
  - Elasticity between -0.8 and -1.6 (instead of -0.1 and -0.8)
- Ioannides and Zabel (2008)
  - Use similar approach, but find no noticeable difference in demand elasticity compared to specification that does not control for neighborhood choice